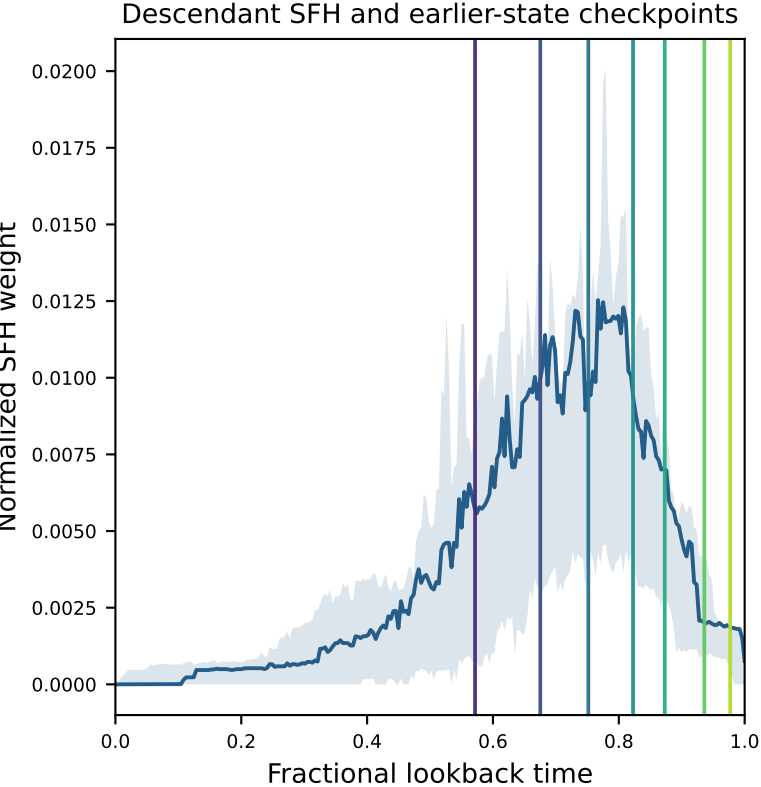
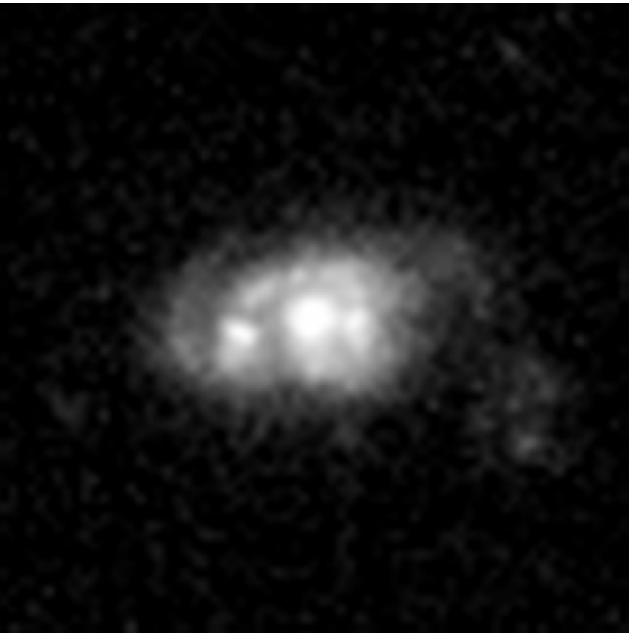
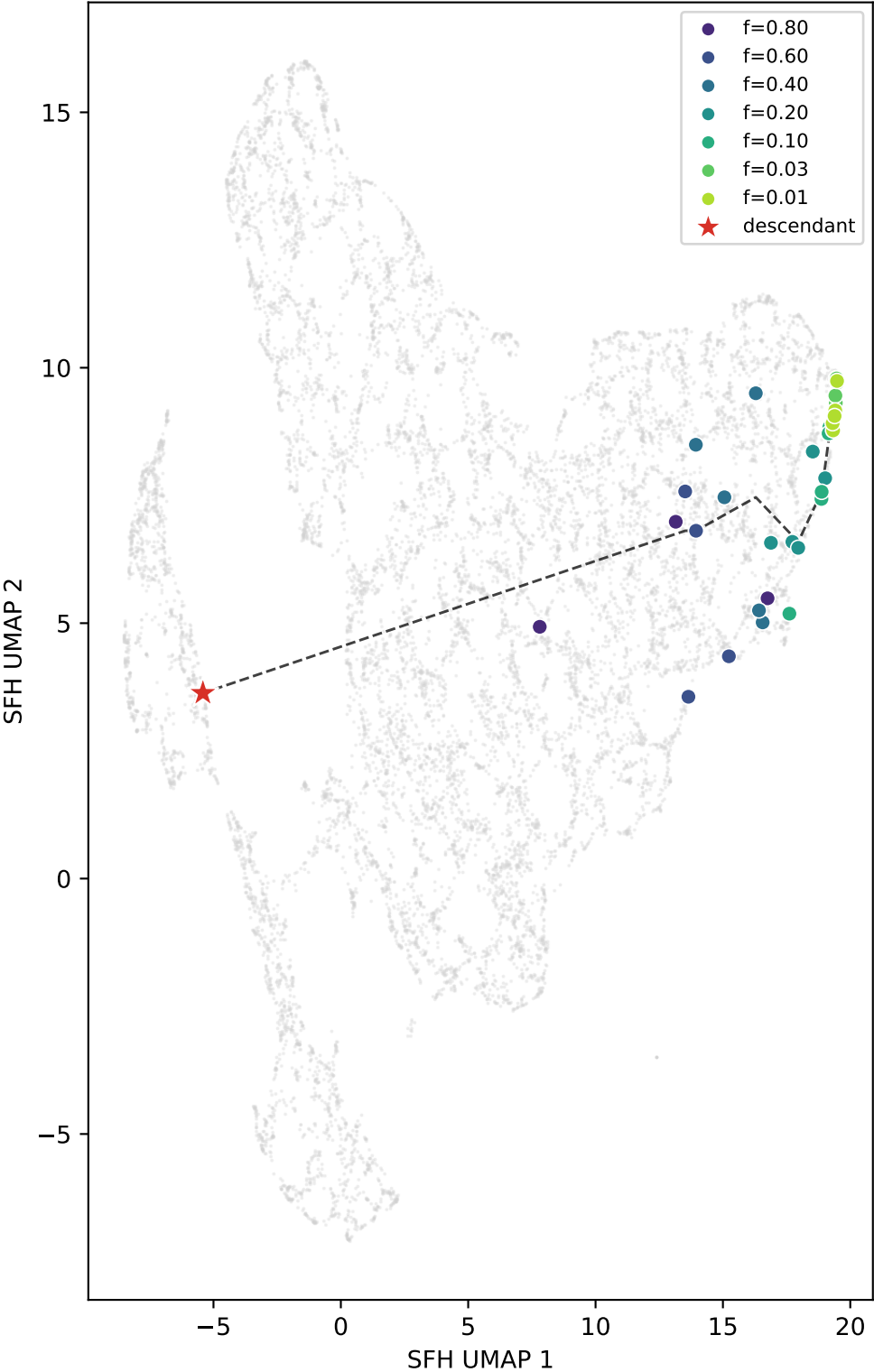


Descendant 2648419808652582361
log M*=11.00, z=0.810



Pilot SFH-selected progenitor-analogue sequence

Analogue positions after SFH-only selection



f	L [Gyr]	log M*	T [Gyr]	pool	z med
0.80	3.77	10.90	2.00	1,389	0.68
0.60	4.45	10.78	2.00	1,664	0.63
0.40	4.95	10.60	1.64	1,862	0.60
0.20	5.42	10.30	1.17	1,998	0.54
0.10	5.75	10.00	0.84	2,043	0.47
0.03	6.17	9.48	0.42	1,655	0.34
0.01	6.44	9.00	0.15	879	0.23

Selection variables

- predicted stellar mass, within +/-0.15 dex
- cumulative SFH before each descendant state
- 2 Gyr physical-time comparison when available

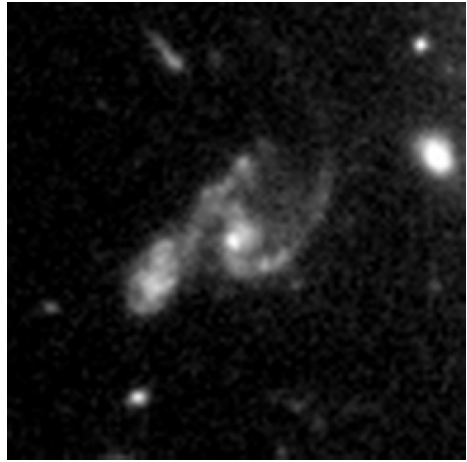
Not used for selection

- candidate redshift
- morphology or image embedding
- UMAP position or present-day sSFR

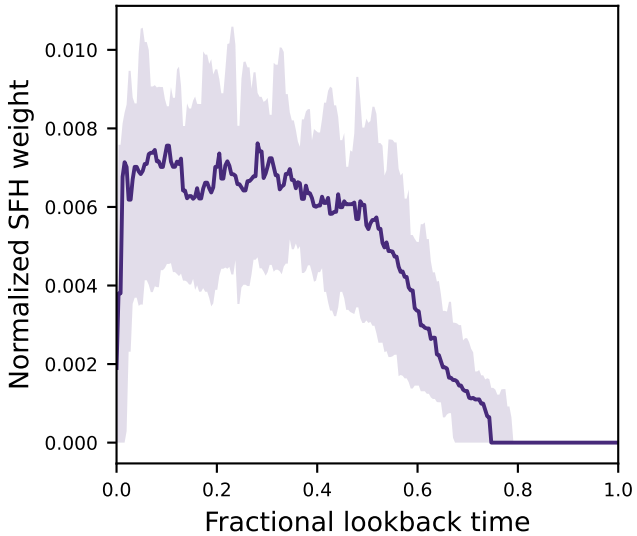
The dashed UMAP path joins median analogue positions. It is an ensemble trend, not a reconstructed orbit of one galaxy.

Earlier state $f=0.80$: lookback=3.77 Gyr, predicted $\log M^*=10.90$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched

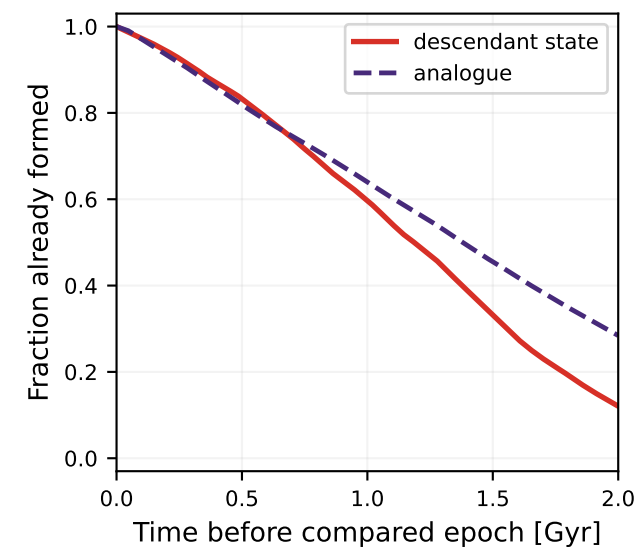
rank 1: 2678711556674304258
 $\log M^*=10.83$ (-0.07) $z=1.334$
 $D=0.065$



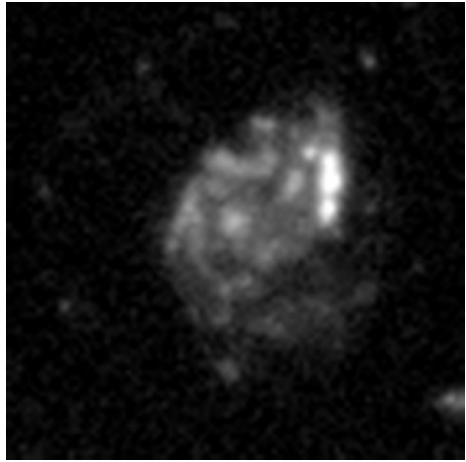
Complete candidate SFH



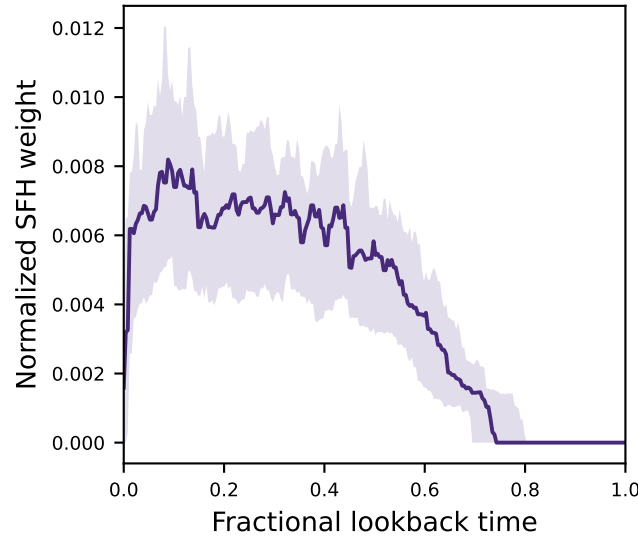
Fractional lookback time



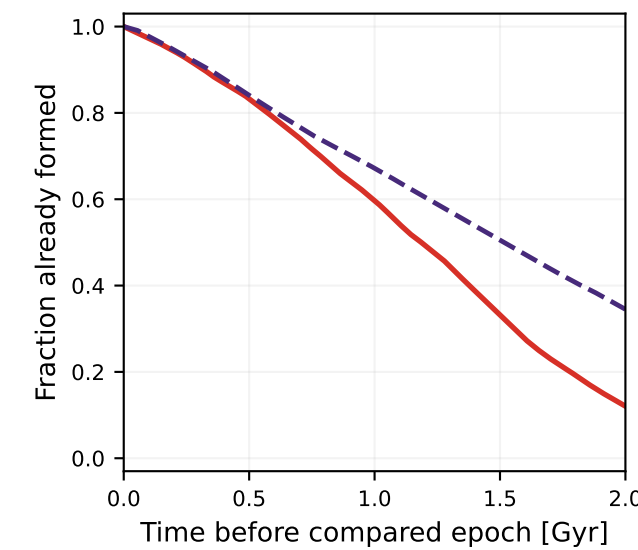
rank 2: 2703488790670980759
 $\log M^*=10.80$ (-0.11) $z=1.170$
 $D=0.093$



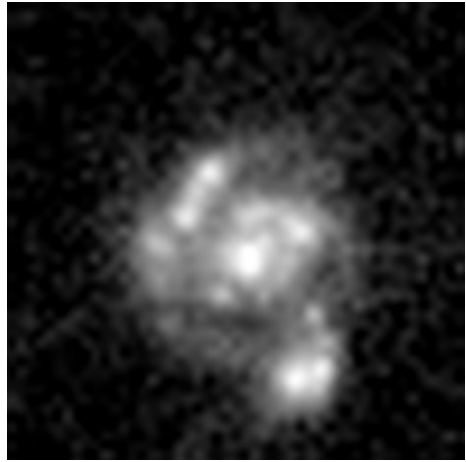
Complete candidate SFH



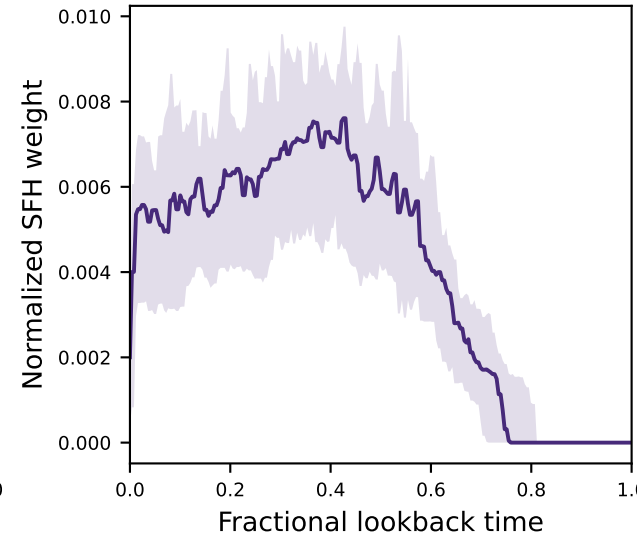
Fractional lookback time



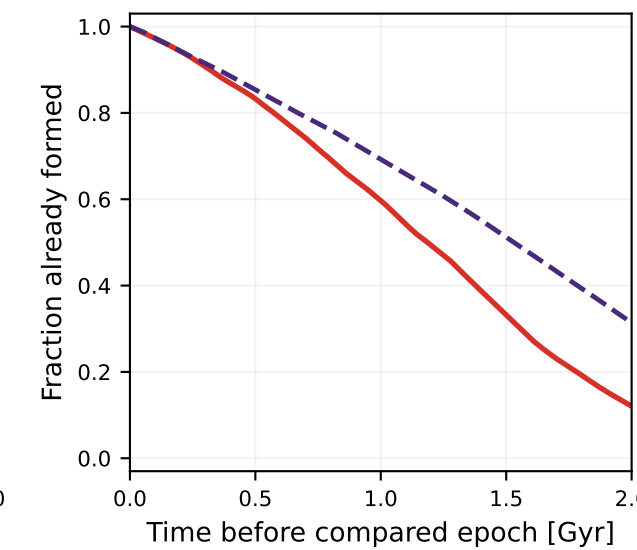
rank 3: 2659854720684798410
 $\log M^*=10.93$ (+0.03) $z=1.378$
 $D=0.101$



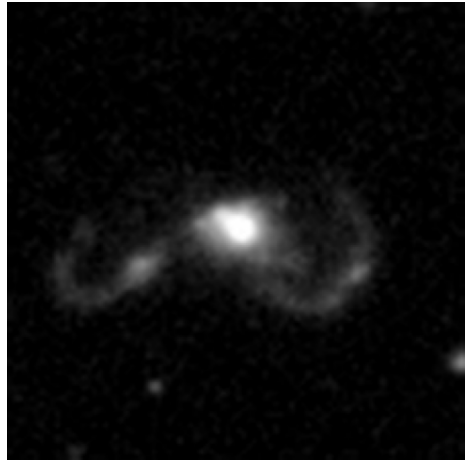
Complete candidate SFH



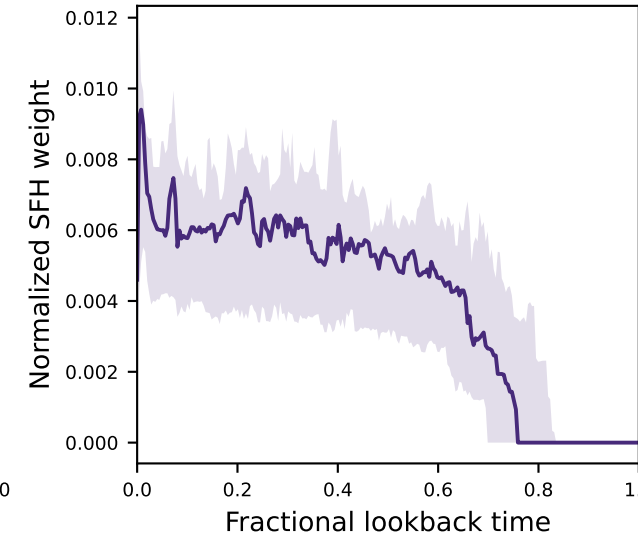
Fractional lookback time



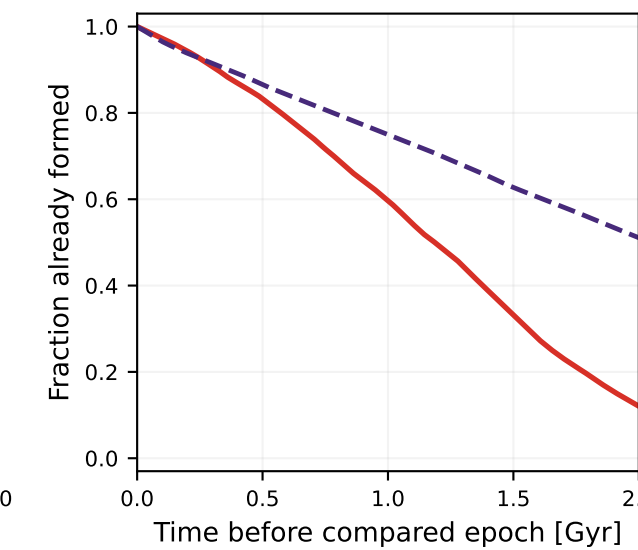
rank 4: 2649173403675031662
 $\log M^*=10.77$ (-0.14) $z=0.832$
 $D=0.170$



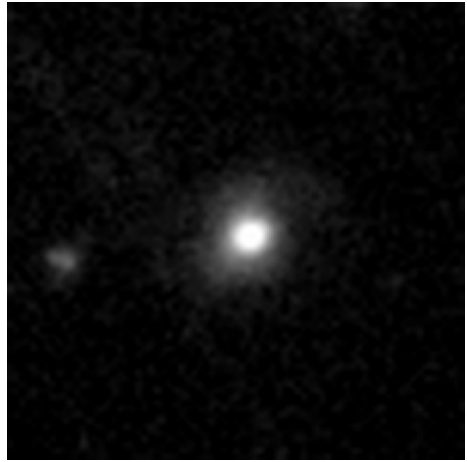
Complete candidate SFH



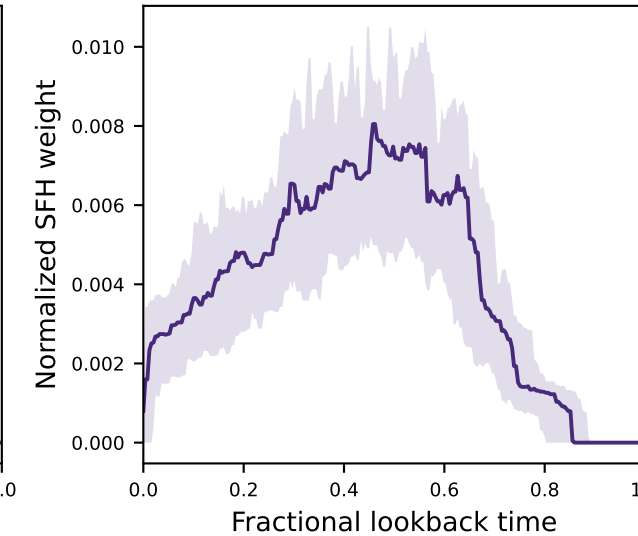
Fractional lookback time



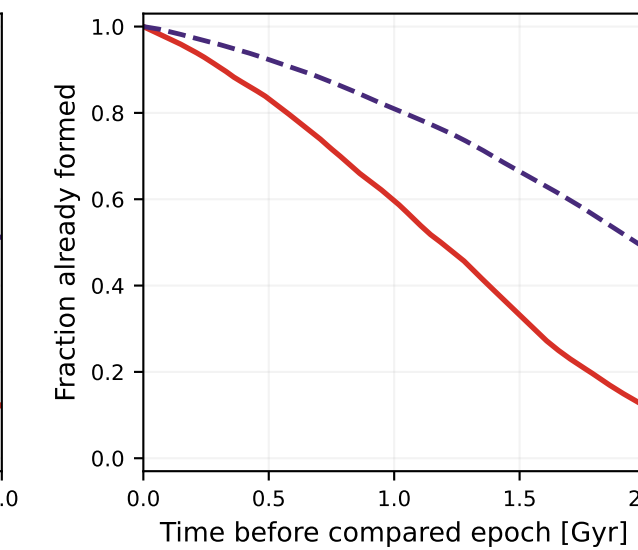
rank 5: 2634119826667500832
 $\log M^*=11.05$ (+0.15) $z=1.336$
 $D=0.209$



Complete candidate SFH

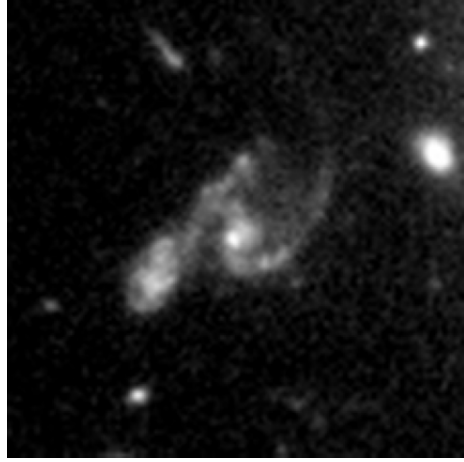


Fractional lookback time

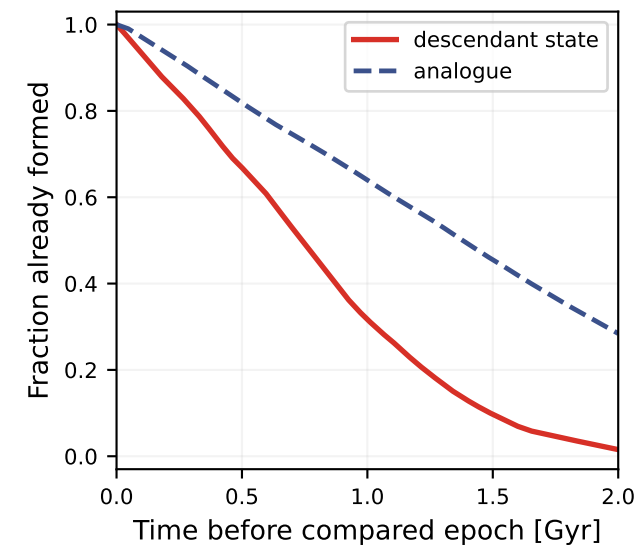
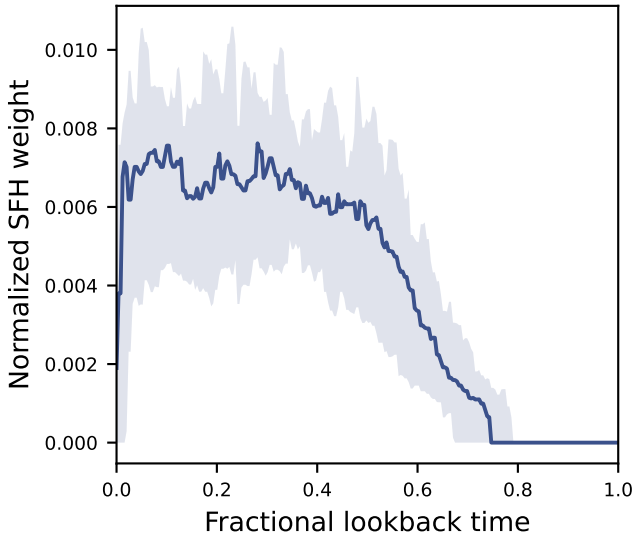


Earlier state $f=0.60$: lookback=4.45 Gyr, predicted $\log M^*=10.78$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched

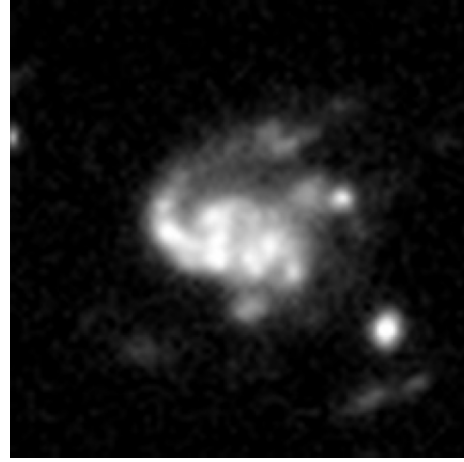
rank 1: 2678711556674304258
 $\log M^*=10.83 (+0.05)$ $z=1.334$
 $D=0.246$



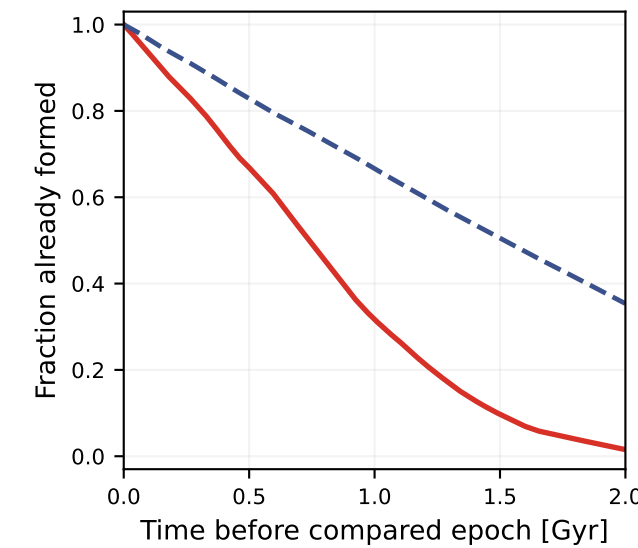
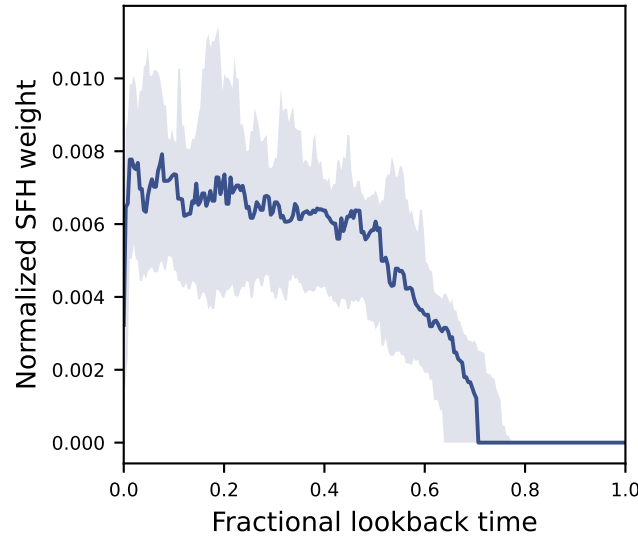
Complete candidate SFH



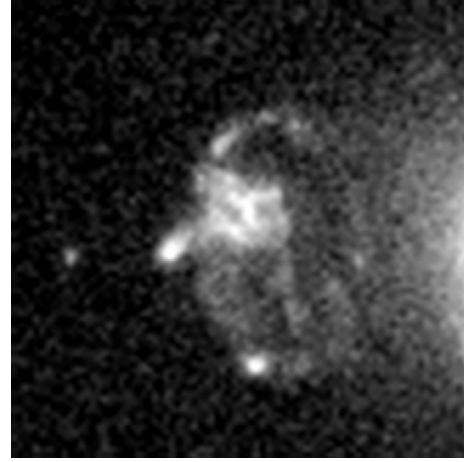
rank 2: 2741447642657085578
 $\log M^*=10.70 (-0.08)$ $z=1.154$
 $D=0.275$



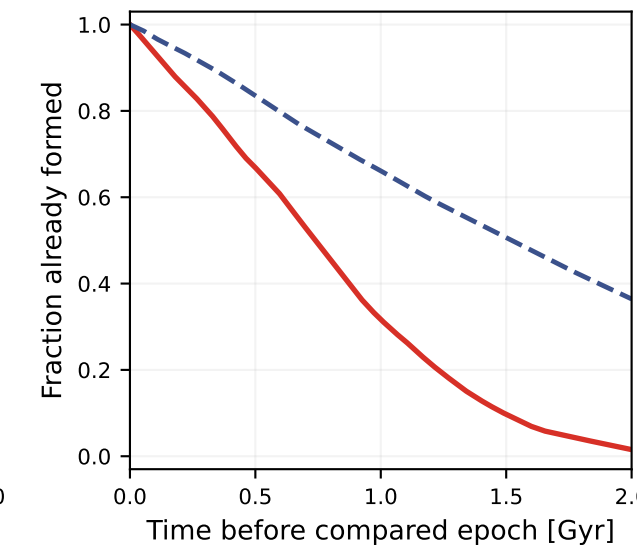
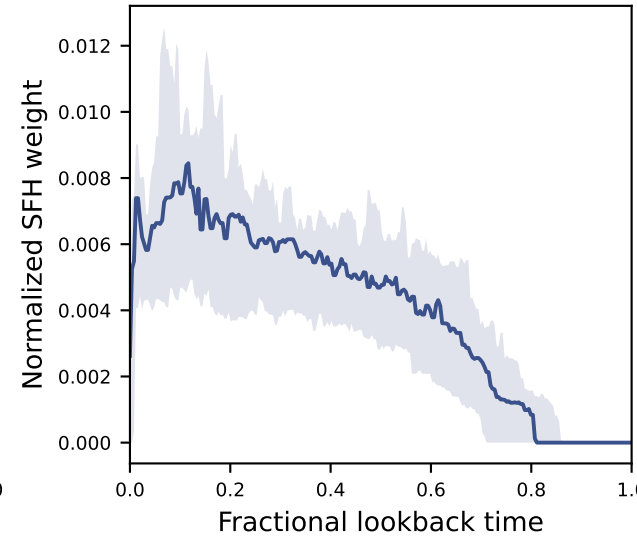
Complete candidate SFH



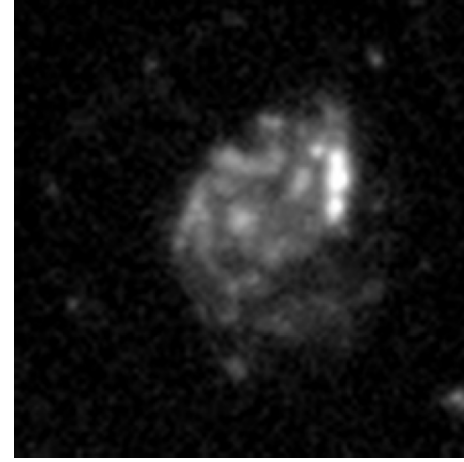
rank 3: 2629425863659052113
 $\log M^*=10.63 (-0.15)$ $z=1.184$
 $D=0.277$



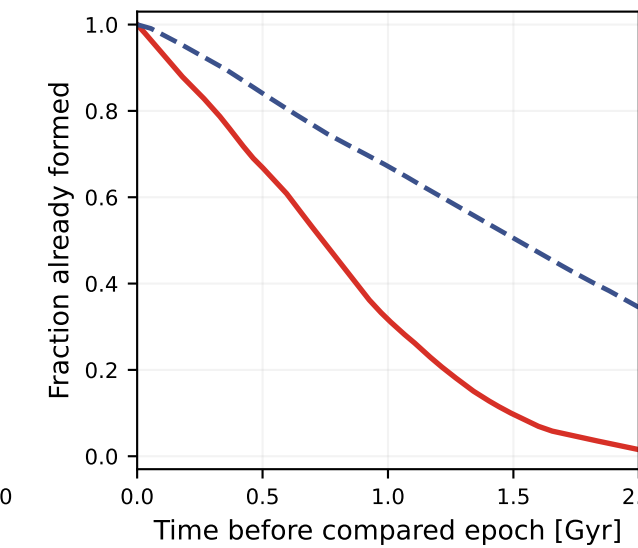
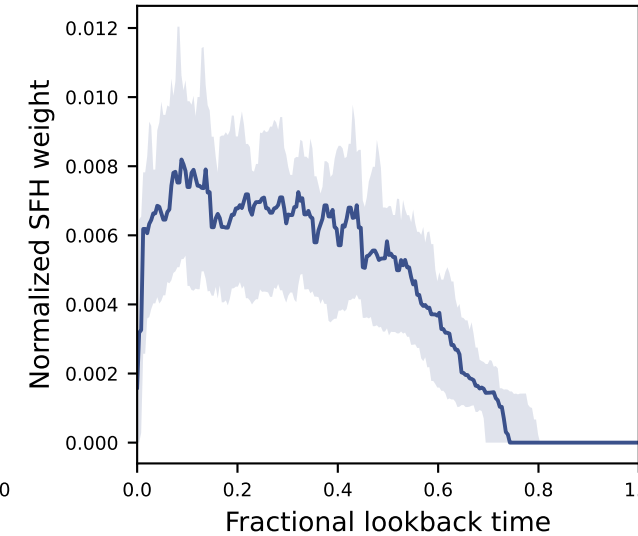
Complete candidate SFH



rank 4: 2703488790670980759
 $\log M^*=10.80 (+0.02)$ $z=1.170$
 $D=0.280$



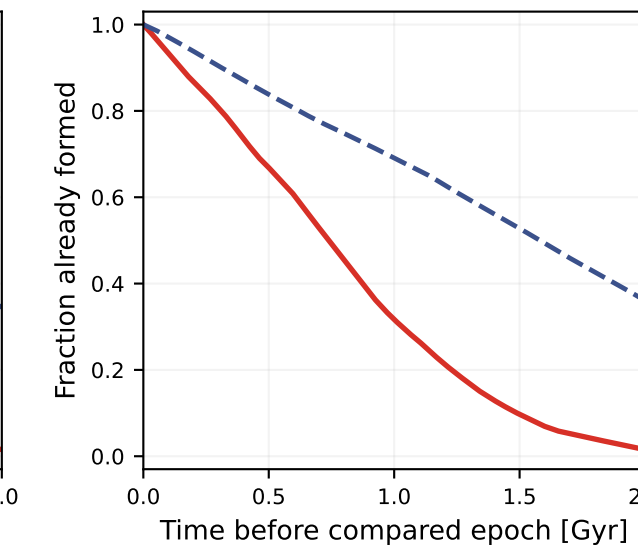
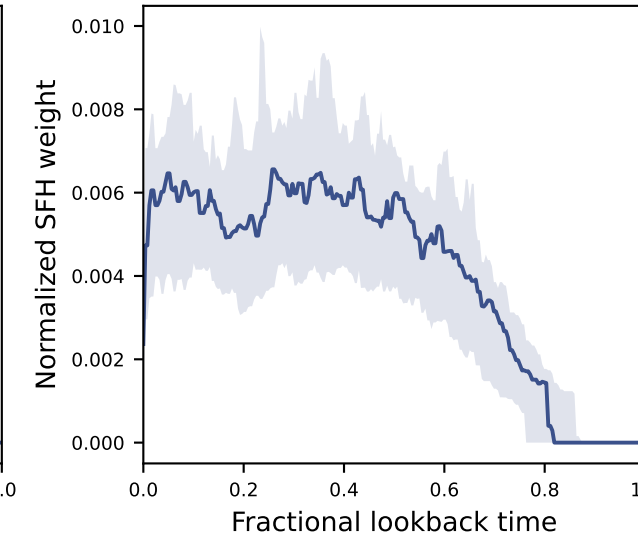
Complete candidate SFH



rank 5: 2652718793642424678
 $\log M^*=10.69 (-0.09)$ $z=1.370$
 $D=0.291$

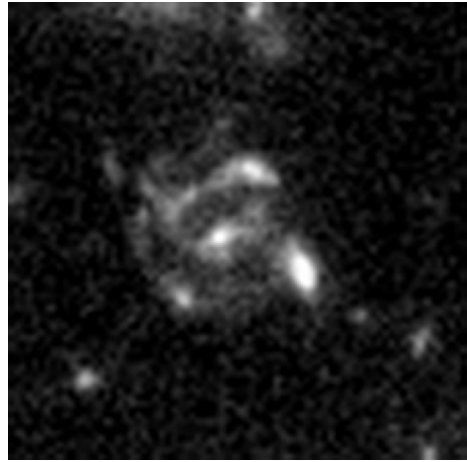


Complete candidate SFH

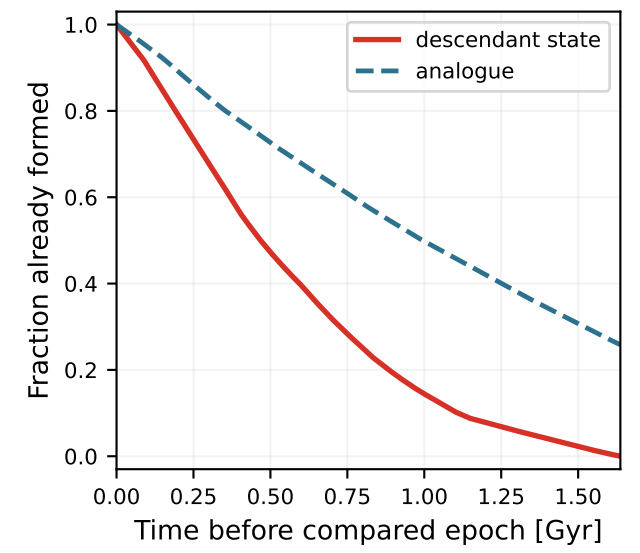
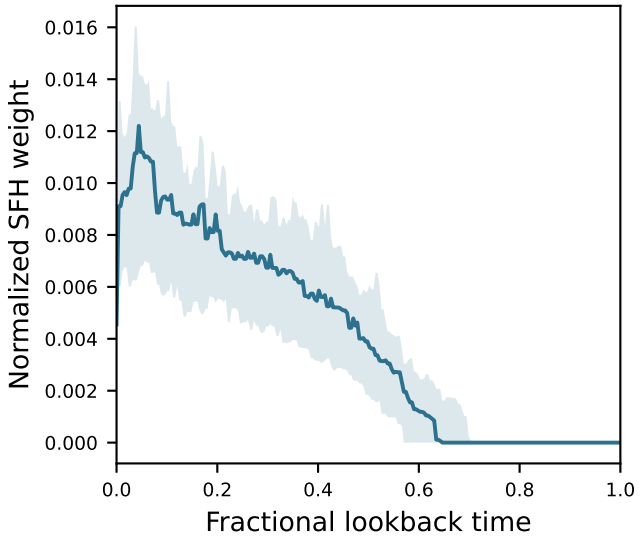


Earlier state $f=0.40$: lookback=4.95 Gyr, predicted $\log M^*=10.60$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched

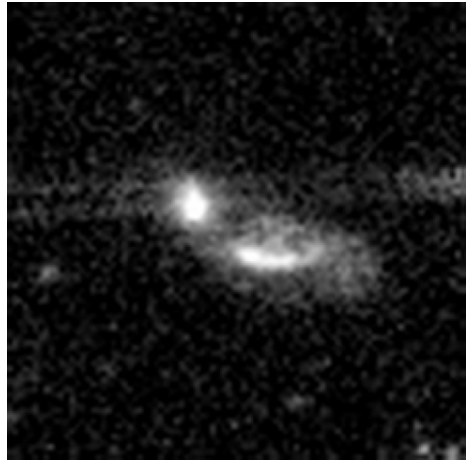
rank 1: 2716117201644464606
 $\log M^*=10.58$ (-0.02) $z=1.350$
 $D=0.258$



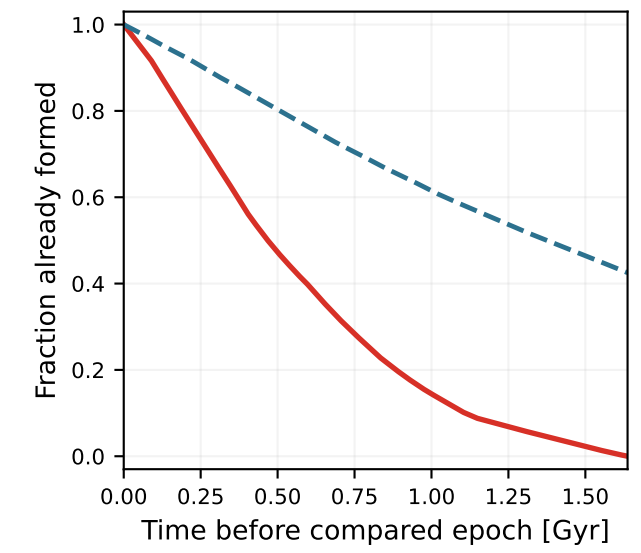
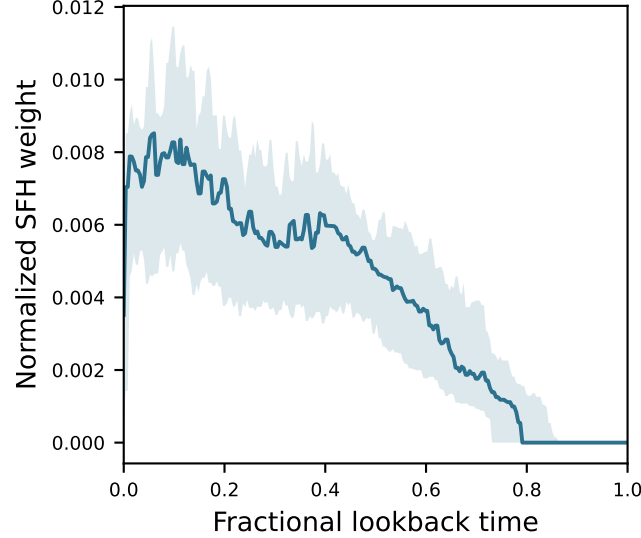
Complete candidate SFH



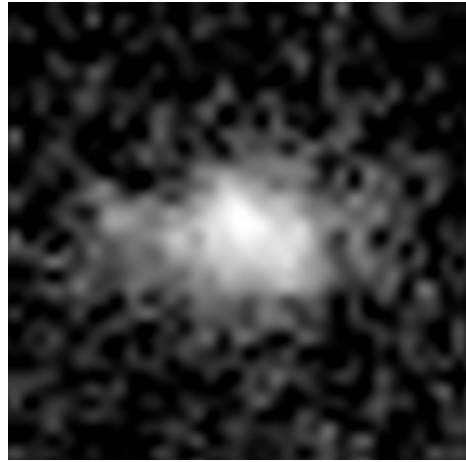
rank 2: 2754251518642661053
 $\log M^*=10.53$ (-0.07) $z=1.250$
 $D=0.356$



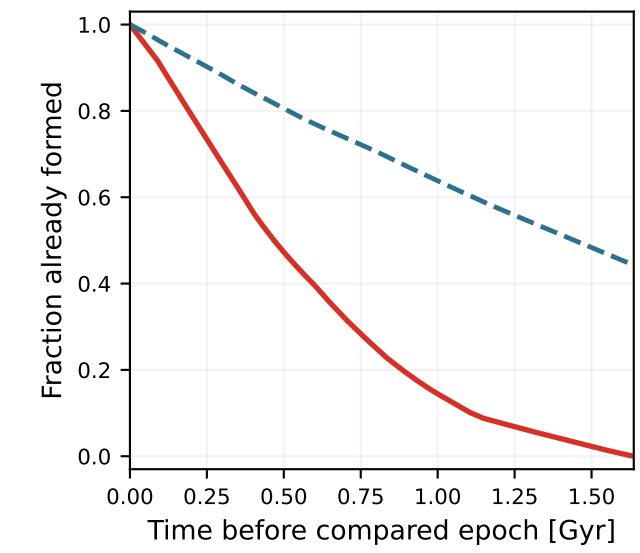
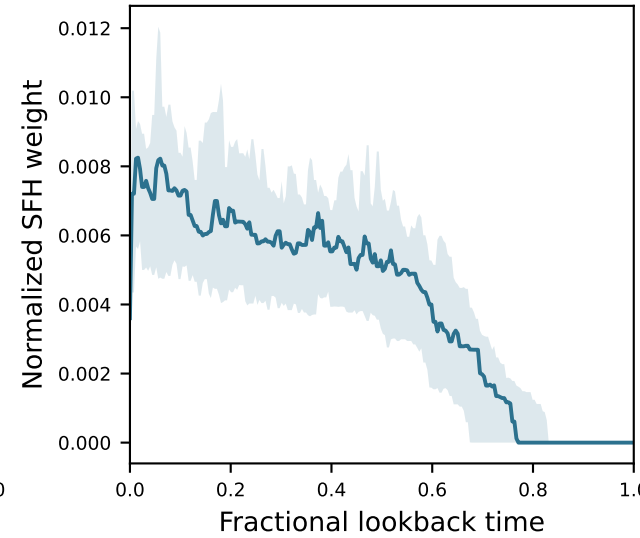
Complete candidate SFH



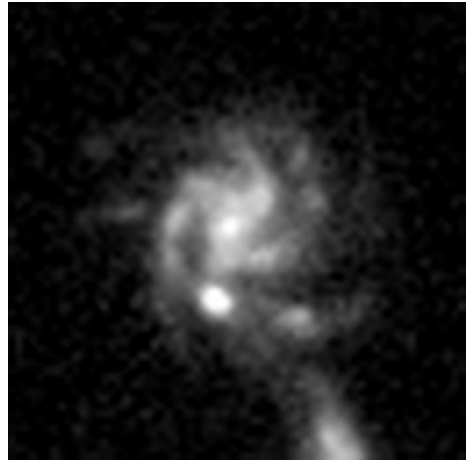
rank 3: 2748216818639415066
 $\log M^*=10.54$ (-0.06) $z=1.266$
 $D=0.368$



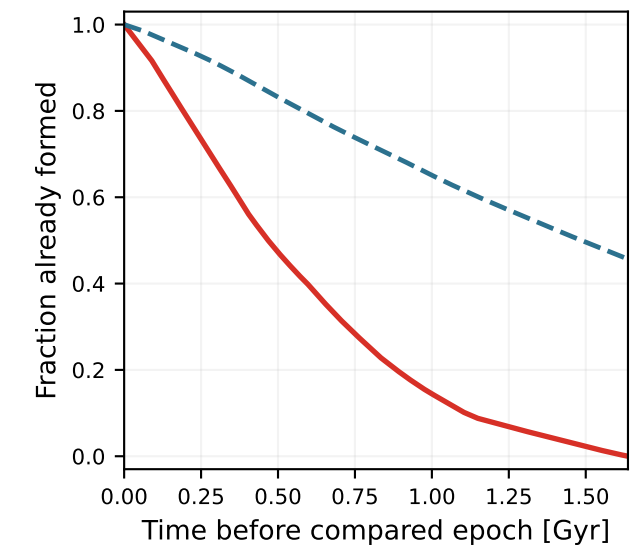
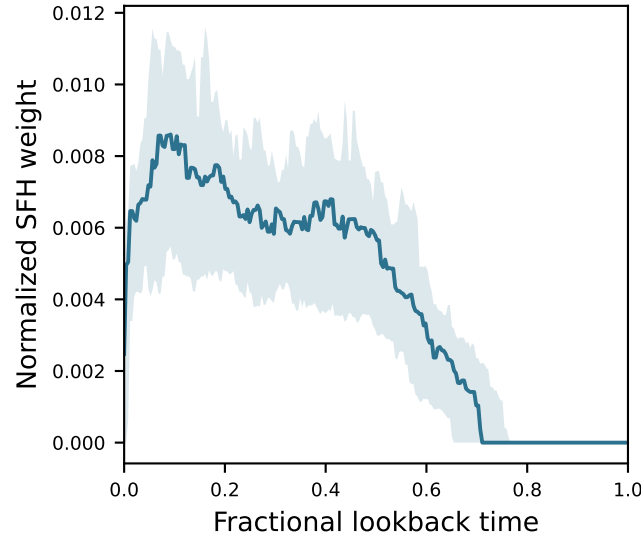
Complete candidate SFH



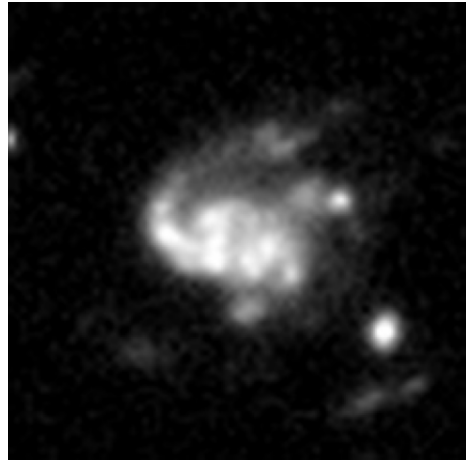
rank 4: 2674024524656252959
 $\log M^*=10.49$ (-0.11) $z=1.118$
 $D=0.385$



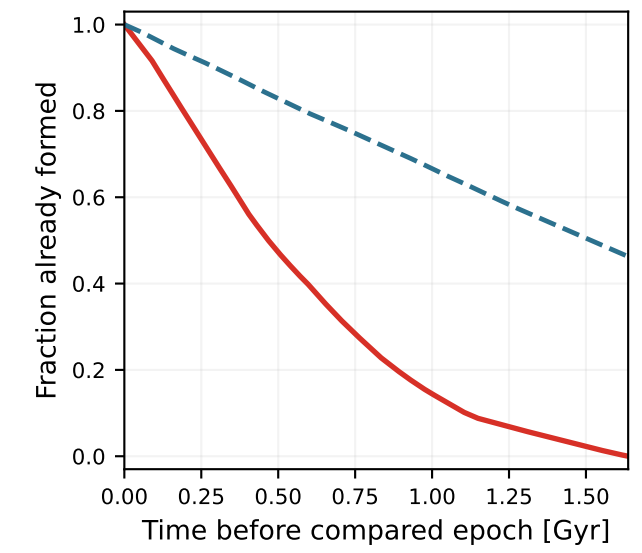
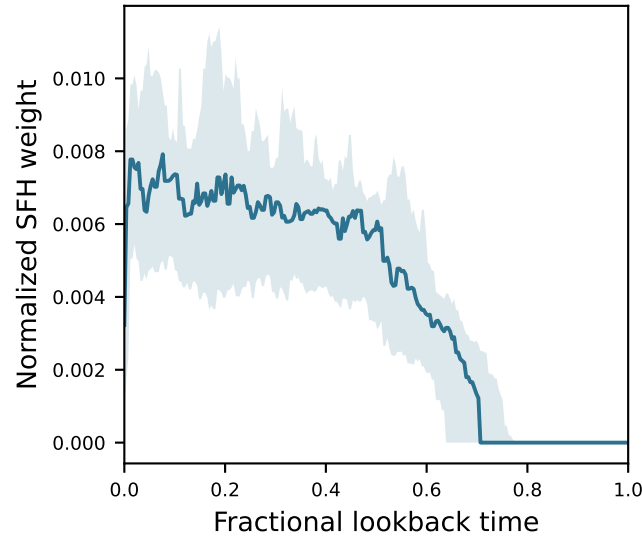
Complete candidate SFH



rank 5: 2741447642657085578
 $\log M^*=10.70$ (+0.10) $z=1.154$
 $D=0.389$

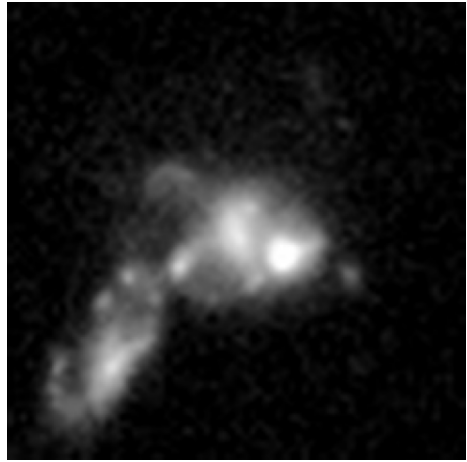


Complete candidate SFH

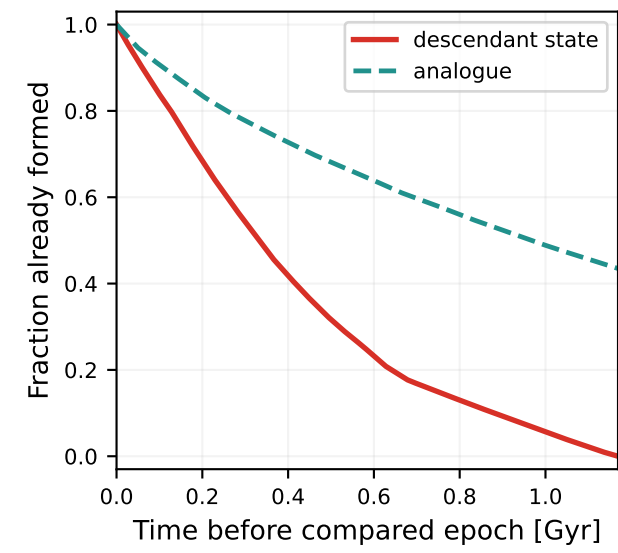
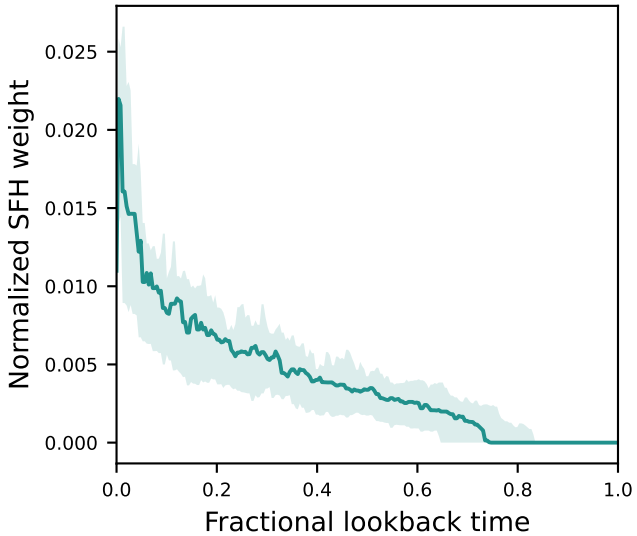


Earlier state $f=0.20$: lookback=5.42 Gyr, predicted $\log M^*=10.30$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched

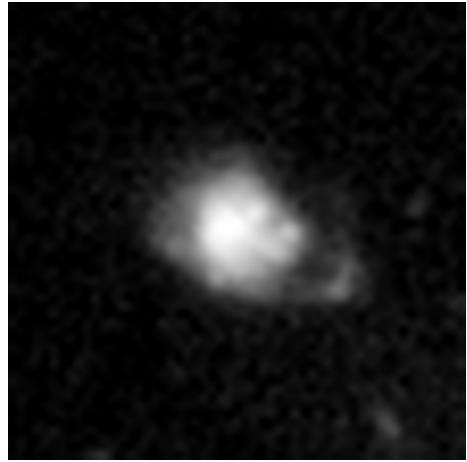
rank 1: 2741023943678283074
 $\log M^*=10.26$ (-0.04) $z=1.194$
 $D=0.323$



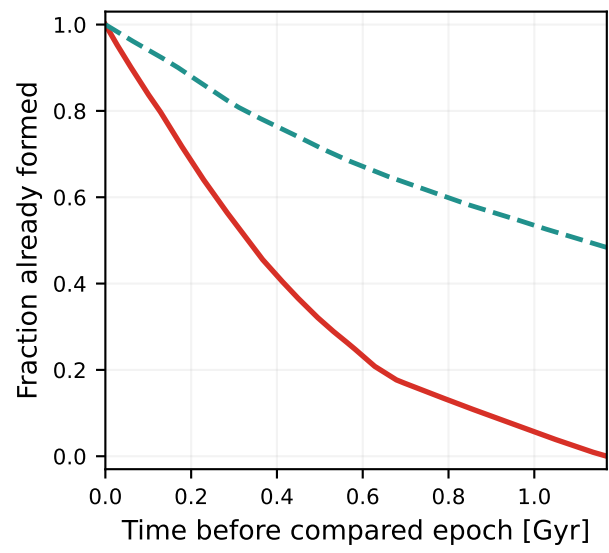
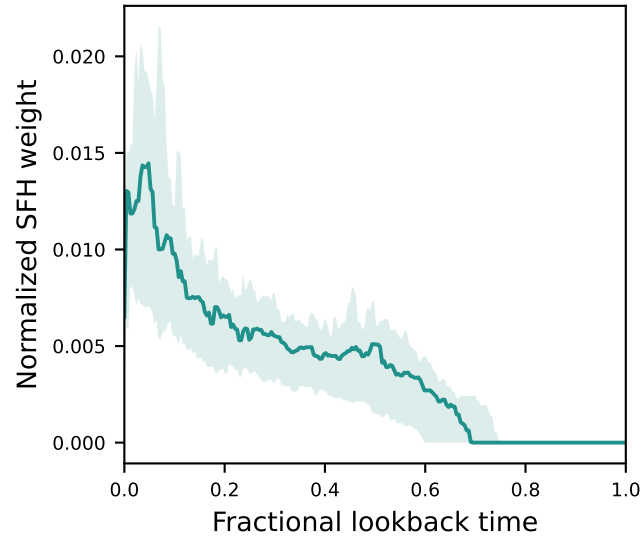
Complete candidate SFH



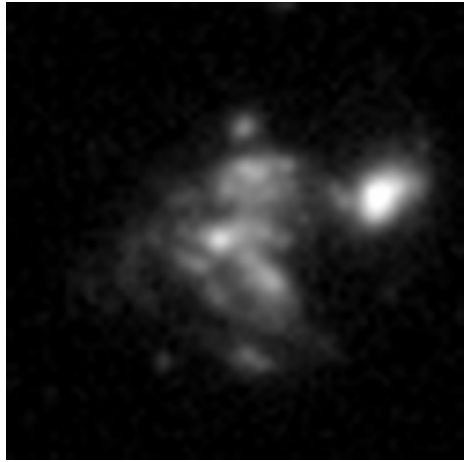
rank 2: 2656685702653947484
 $\log M^*=10.28$ (-0.02) $z=1.122$
 $D=0.361$



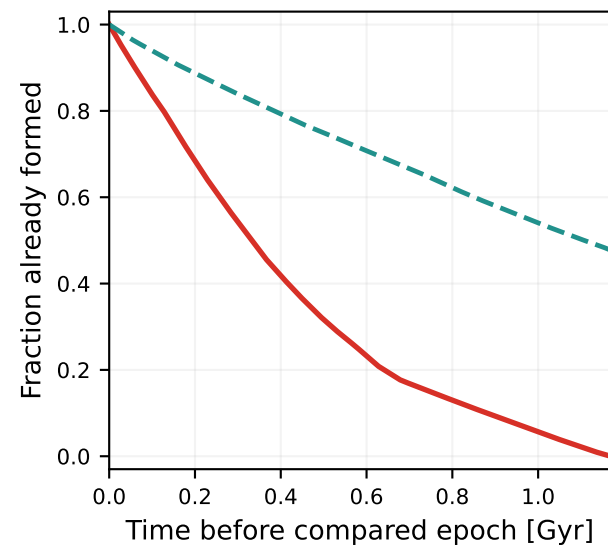
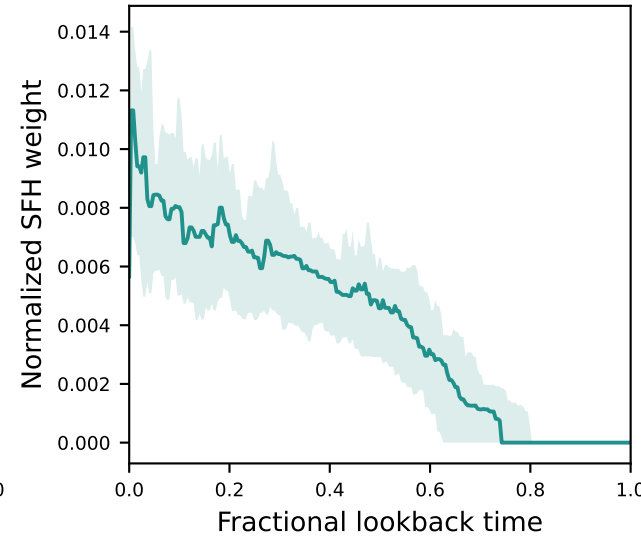
Complete candidate SFH



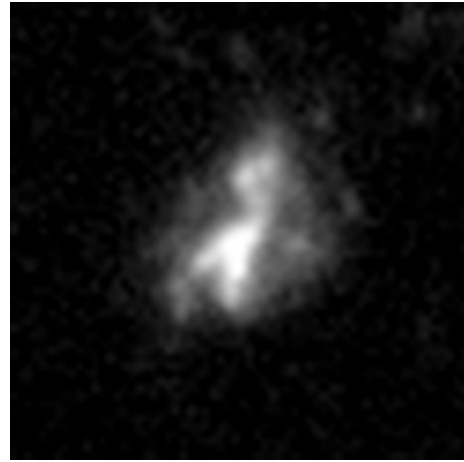
rank 3: 2640109785664925808
 $\log M^*=10.35$ (+0.05) $z=1.478$
 $D=0.378$



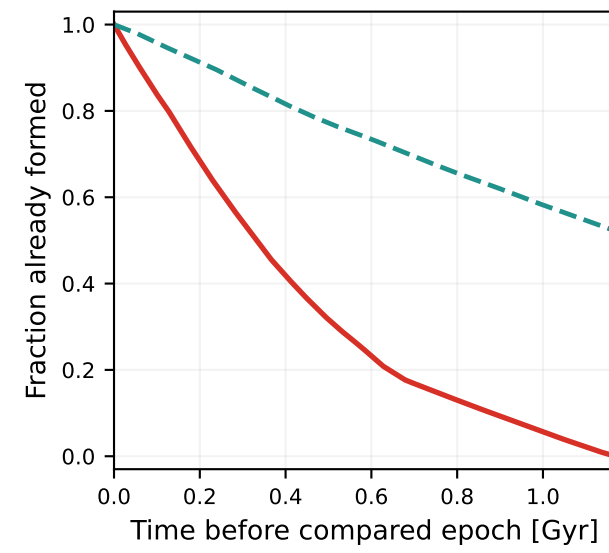
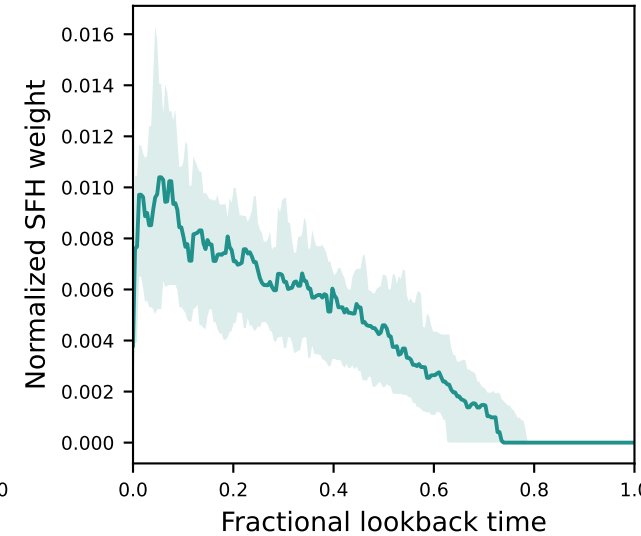
Complete candidate SFH



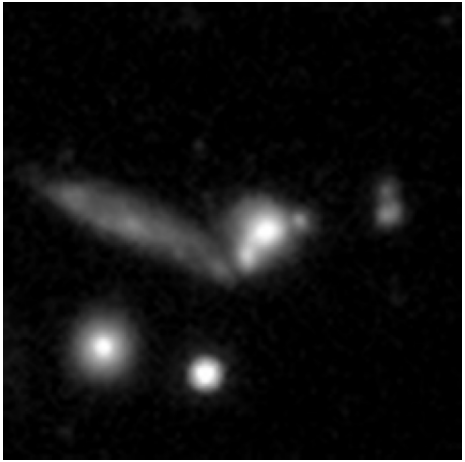
rank 4: 2717353722653902287
 $\log M^*=10.23$ (-0.07) $z=1.198$
 $D=0.407$



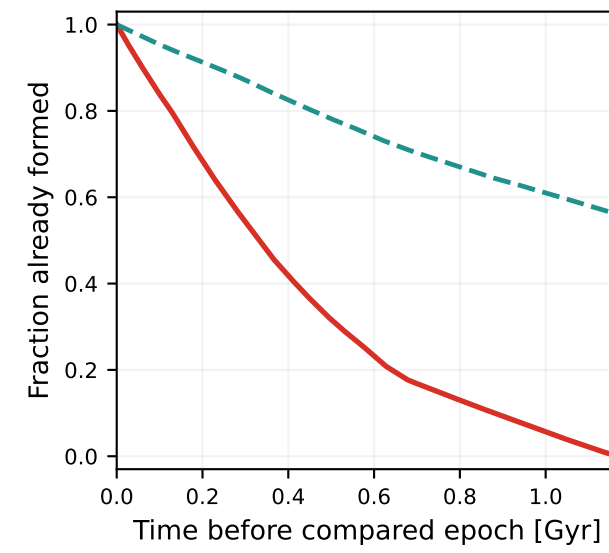
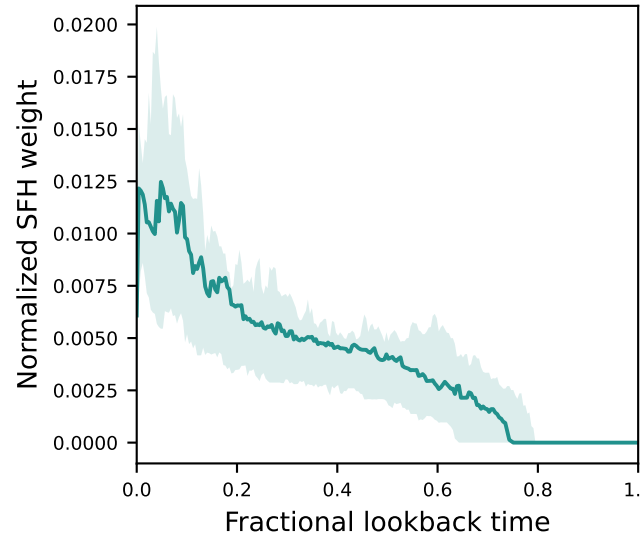
Complete candidate SFH



rank 5: 2653905751651371339
 $\log M^*=10.18$ (-0.12) $z=0.842$
 $D=0.419$

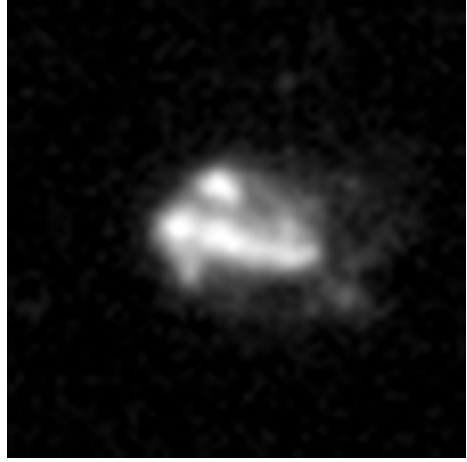


Complete candidate SFH

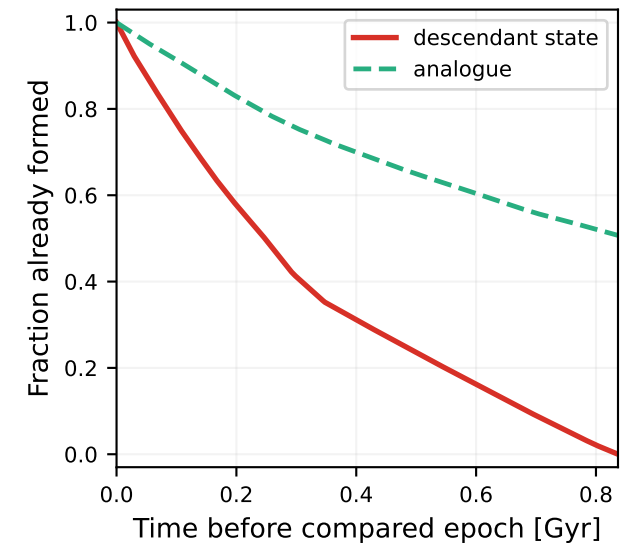
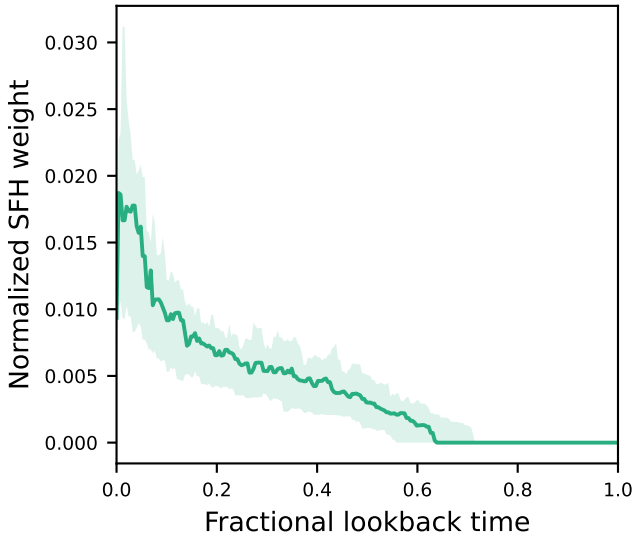


Earlier state $f=0.10$: lookback=5.75 Gyr, predicted $\log M^*=10.00$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched

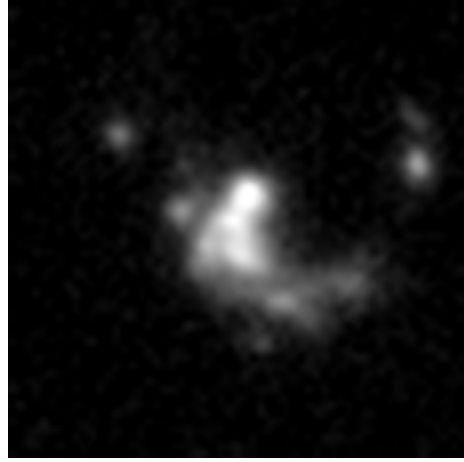
rank 1: 2647131718660160391
 $\log M^*=10.03 (+0.03)$ $z=1.170$
 $D=0.347$



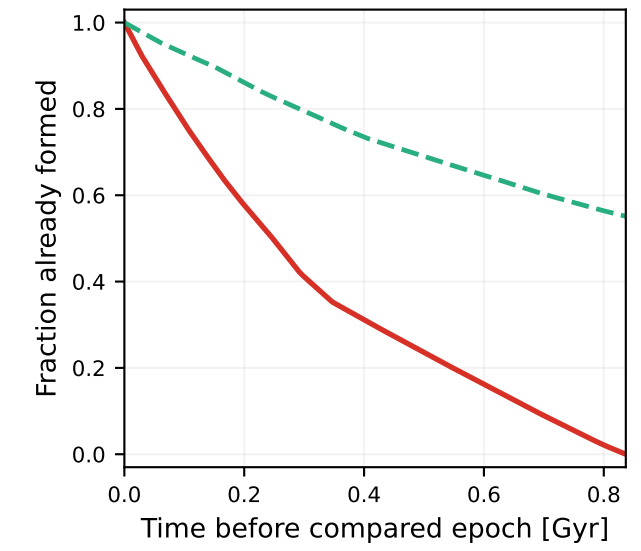
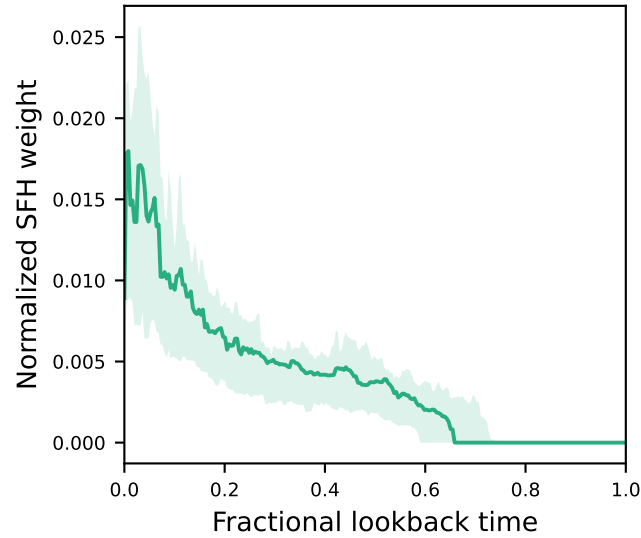
Complete candidate SFH



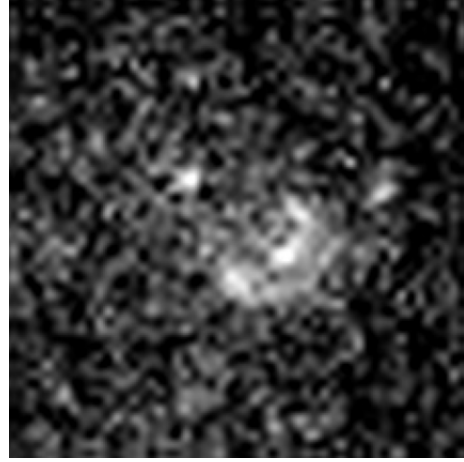
rank 2: 2650234495658951302
 $\log M^*=9.93 (-0.07)$ $z=1.010$
 $D=0.381$



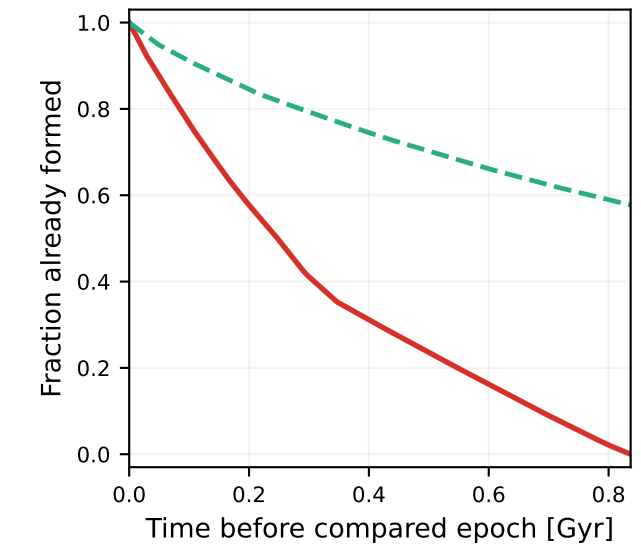
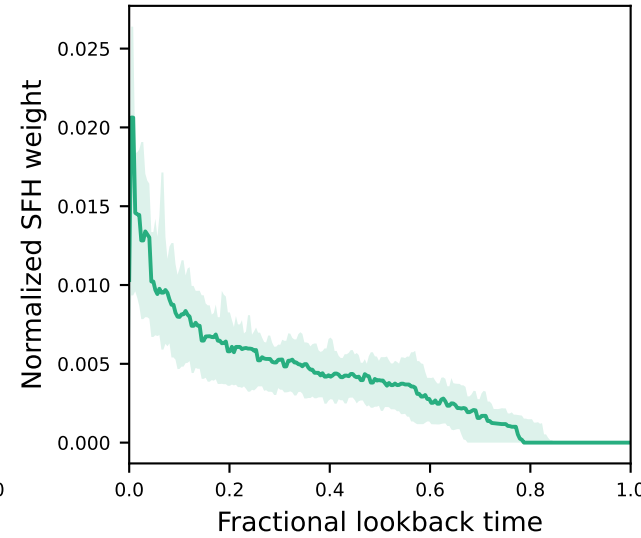
Complete candidate SFH



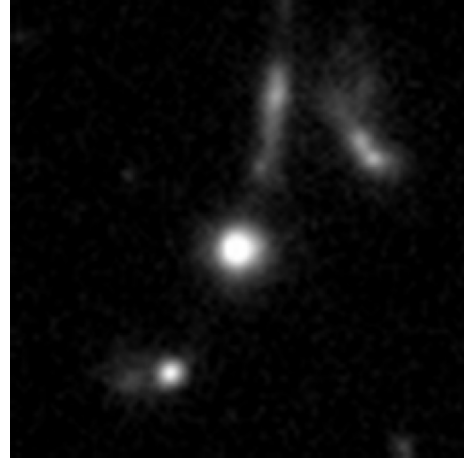
rank 3: 2647176063640000279
 $\log M^*=9.95 (-0.05)$ $z=1.240$
 $D=0.386$



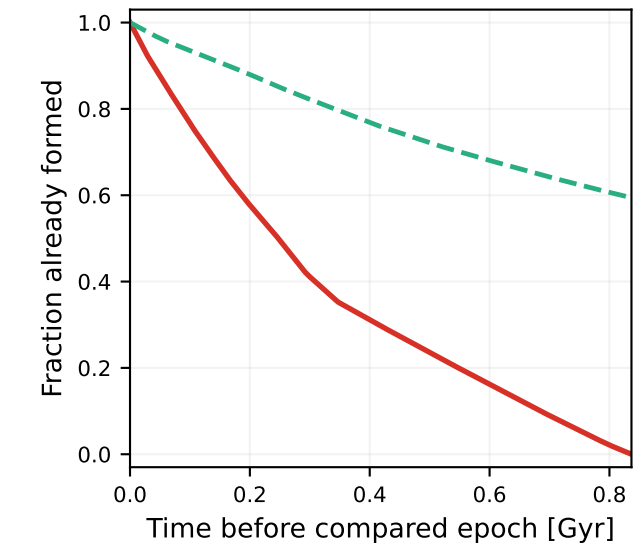
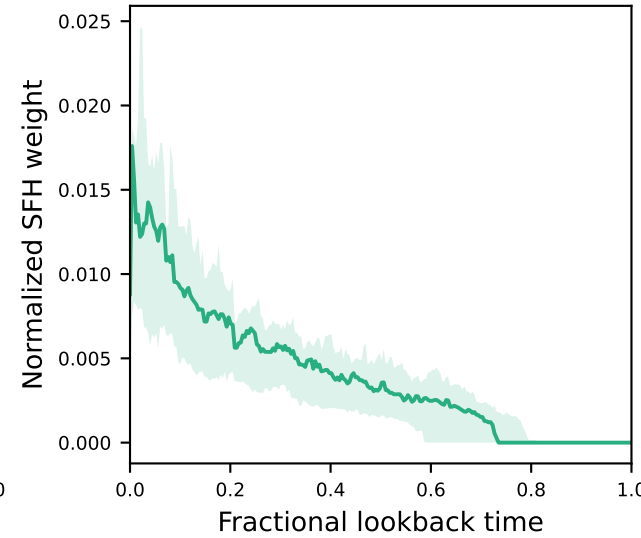
Complete candidate SFH



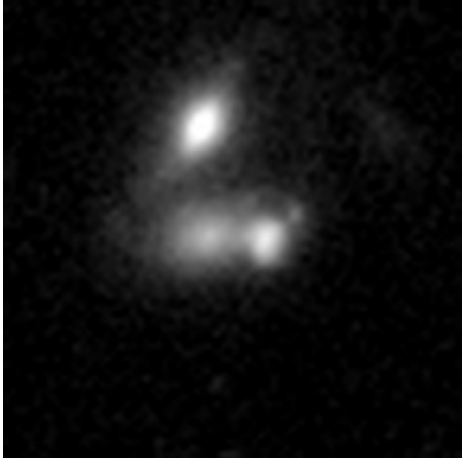
rank 4: 2684931694682266433
 $\log M^*=10.09 (+0.09)$ $z=0.984$
 $D=0.408$



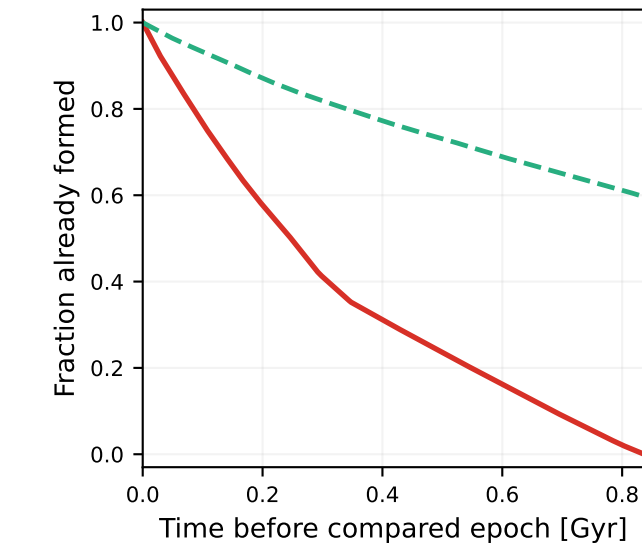
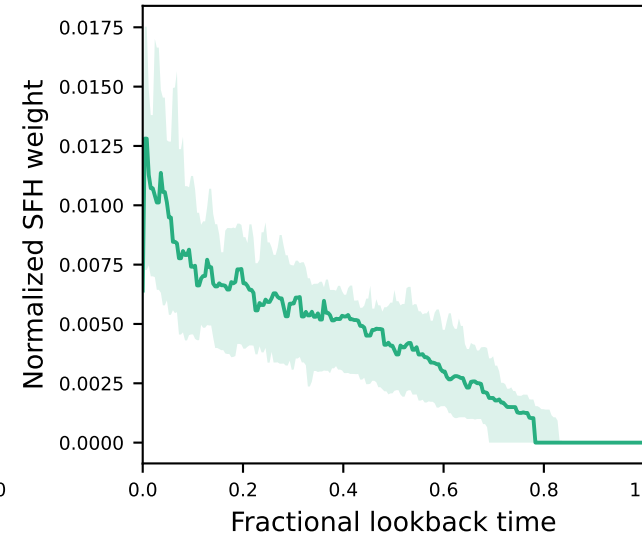
Complete candidate SFH



rank 5: 2658189408677769790
 $\log M^*=10.07 (+0.07)$ $z=1.464$
 $D=0.410$



Complete candidate SFH

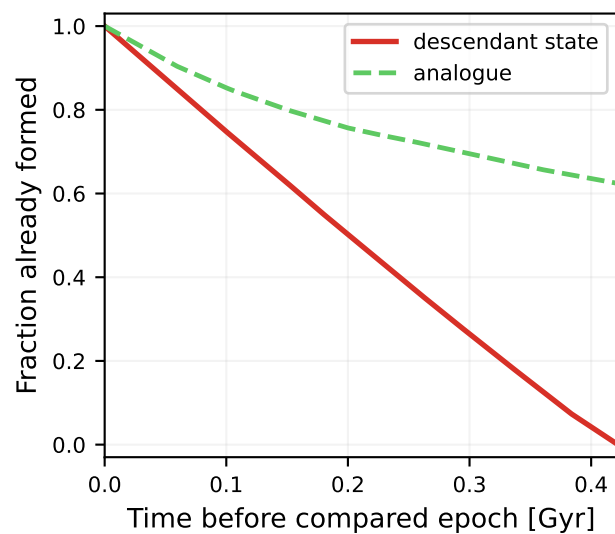
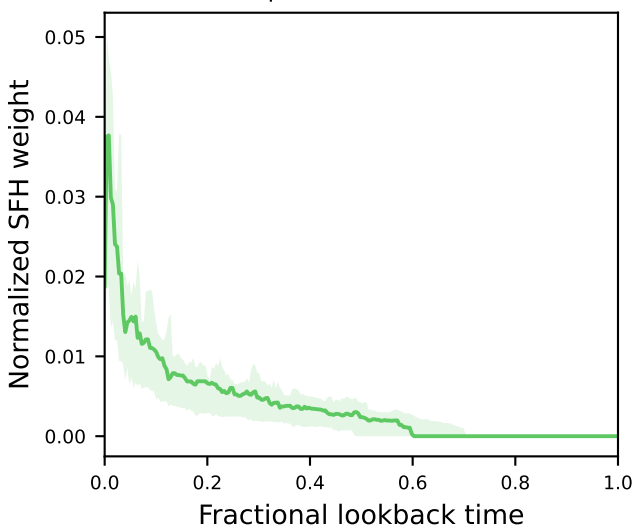


Earlier state $f=0.03$: lookback=6.17 Gyr, predicted $\log M^*=9.48$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 CAUTION: only 0.42 Gyr of pre-state history exists; treat this checkpoint as weakly constrained

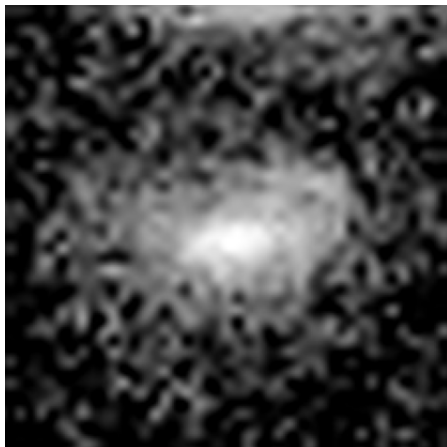
rank 1: 2648545772681764910
 $\log M^*=9.39$ (-0.08) $z=0.994$
 $D=0.288$



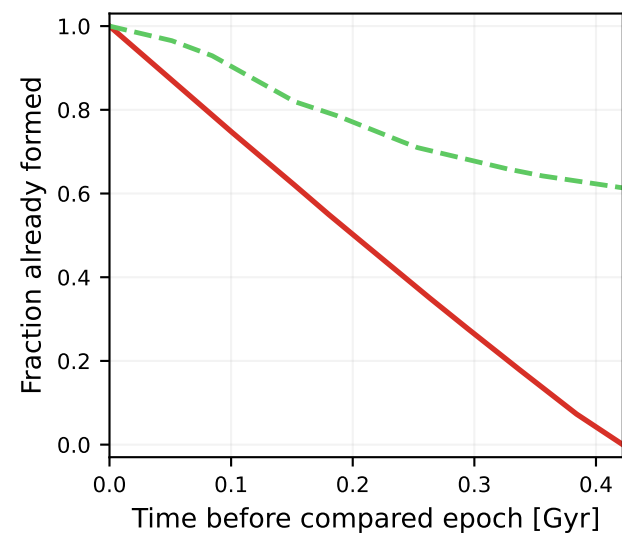
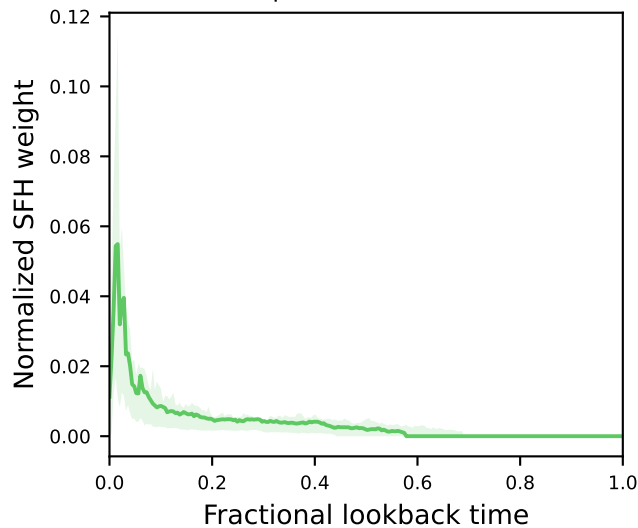
Complete candidate SFH



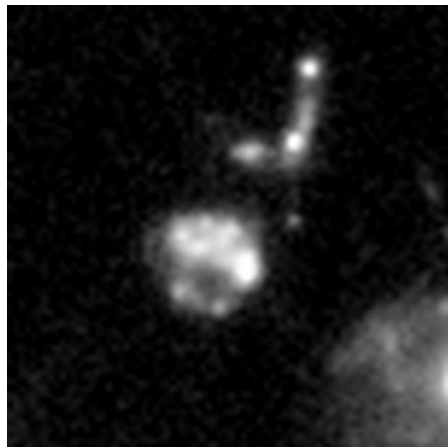
rank 2: 2647975483685285807
 $\log M^*=9.44$ (-0.04) $z=0.510$
 $D=0.298$



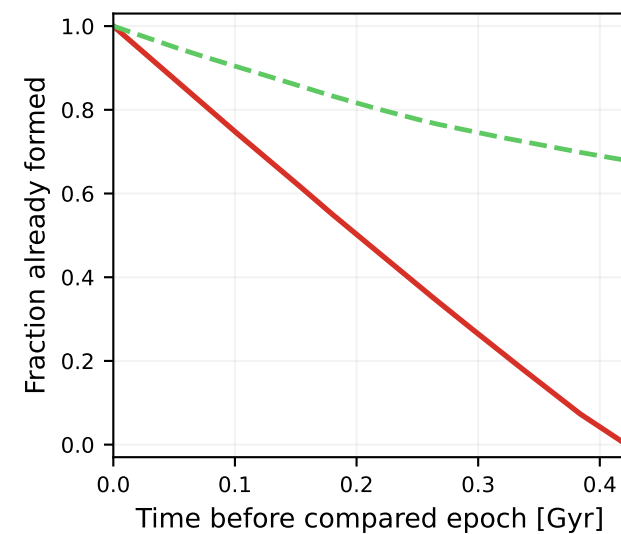
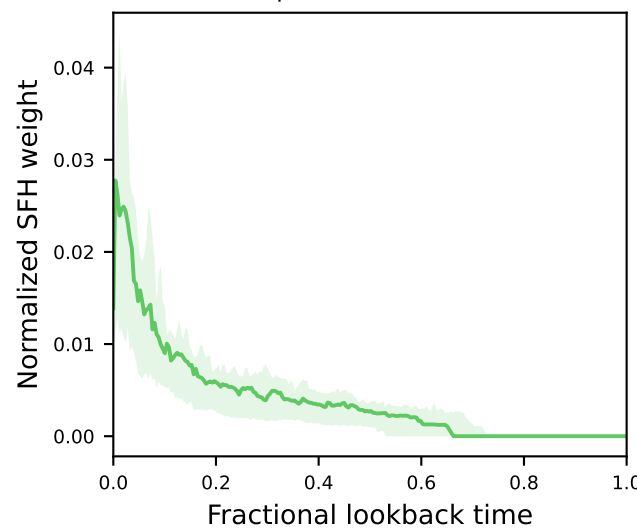
Complete candidate SFH



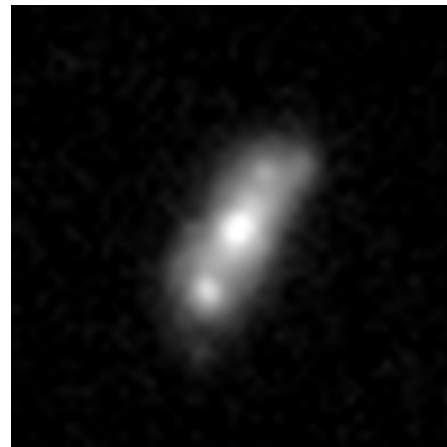
rank 3: 2740268308665100250
 $\log M^*=9.47$ (-0.01) $z=0.760$
 $D=0.337$



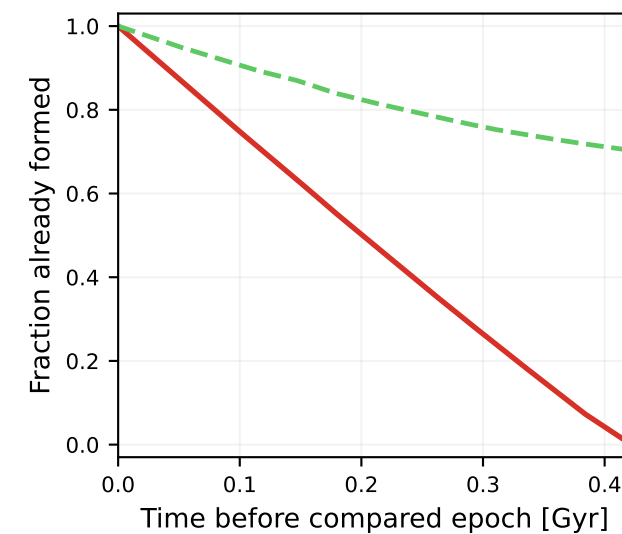
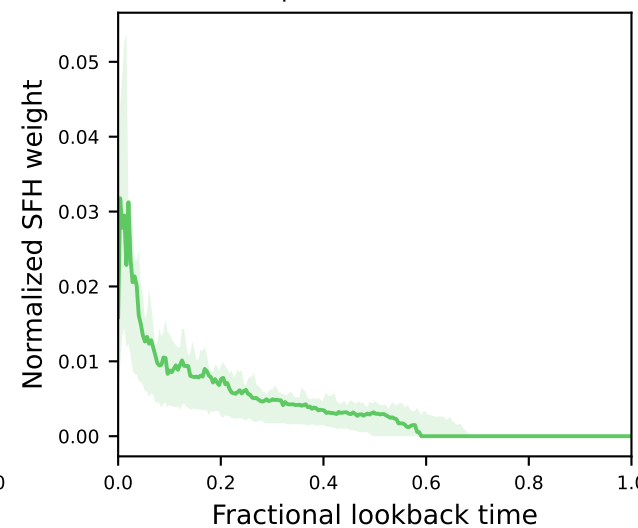
Complete candidate SFH



rank 4: 2753054265674679912
 $\log M^*=9.50$ (+0.02) $z=0.560$
 $D=0.346$



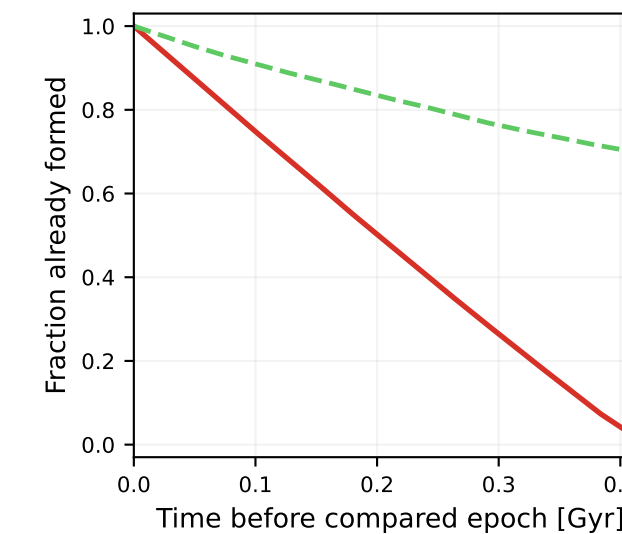
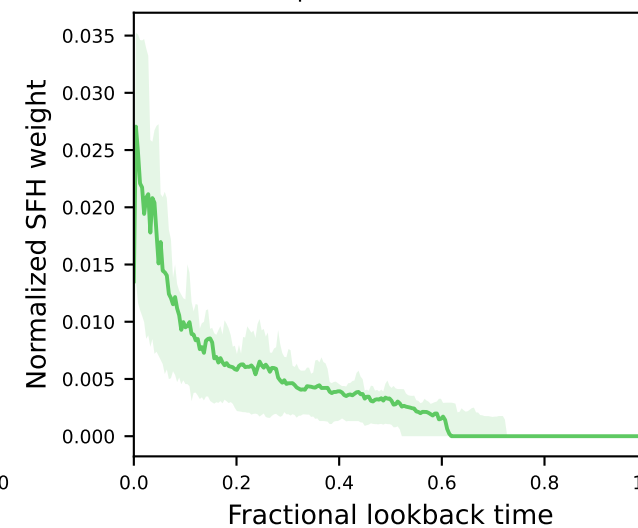
Complete candidate SFH



rank 5: 2729264667640595011
 $\log M^*=9.49$ (+0.01) $z=0.760$
 $D=0.349$

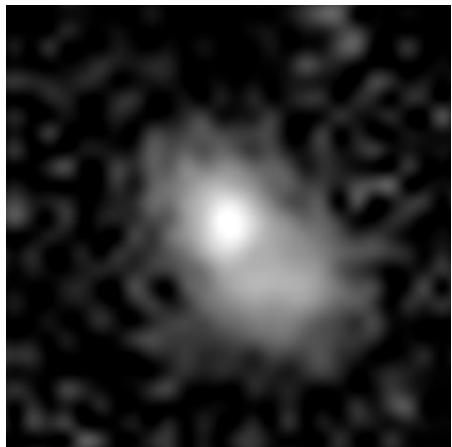


Complete candidate SFH

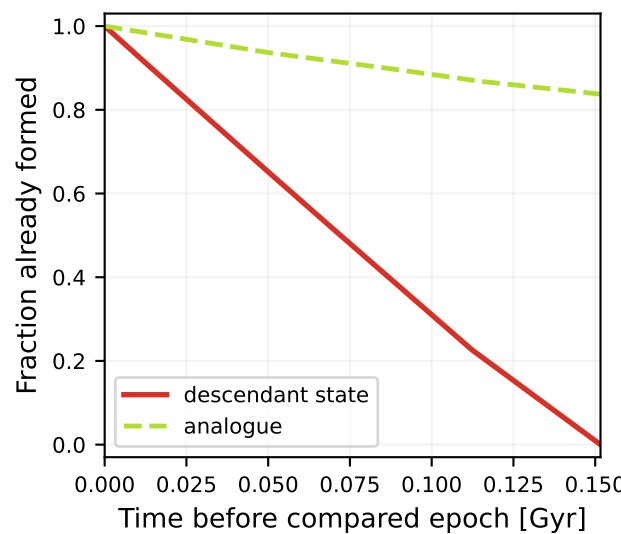
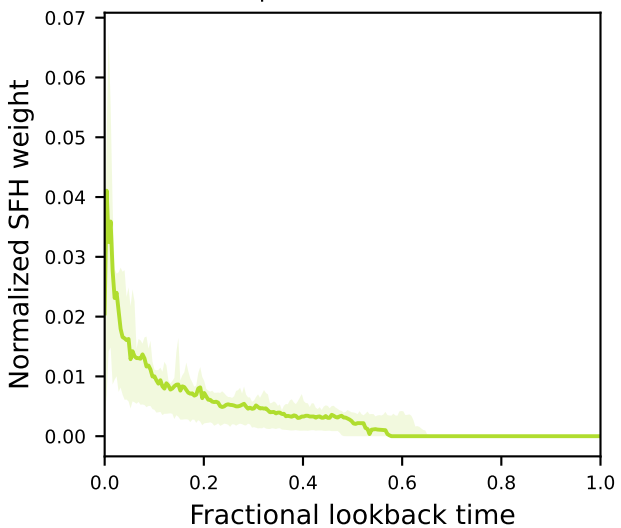


Earlier state $f=0.01$: lookback=6.44 Gyr, predicted $\log M^*=9.00$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 CAUTION: only 0.15 Gyr of pre-state history exists; treat this checkpoint as weakly constrained

rank 1: 2730218645688754837
 $\log M^*=8.88$ (-0.12) $z=0.560$
 $D=0.430$



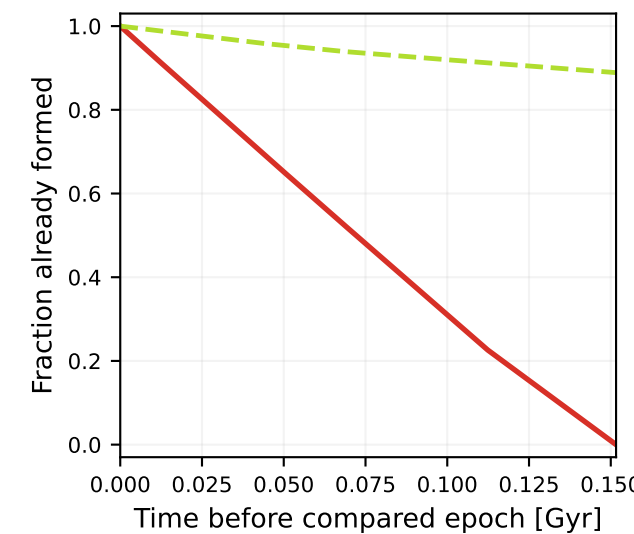
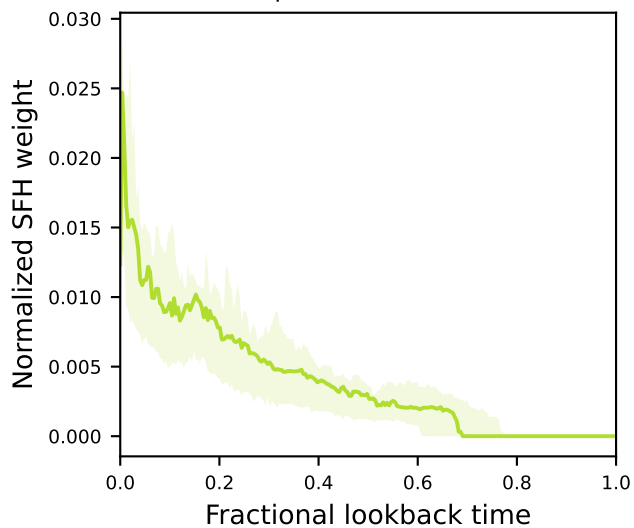
Complete candidate SFH



rank 2: 2674598393649348048
 $\log M^*=9.10$ (+0.10) $z=0.848$
 $D=0.457$



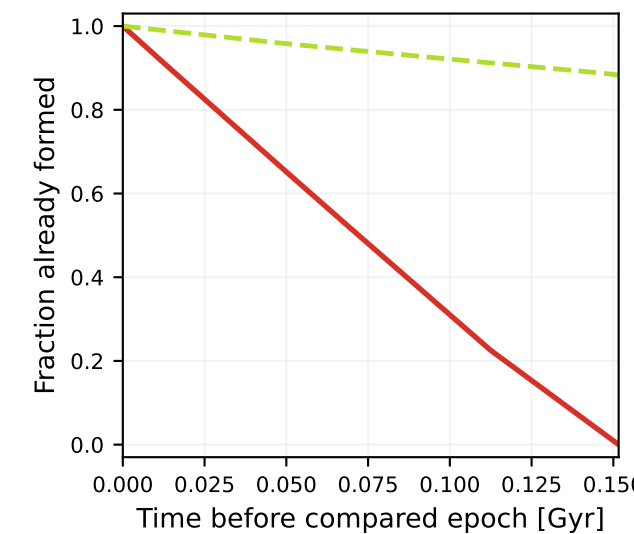
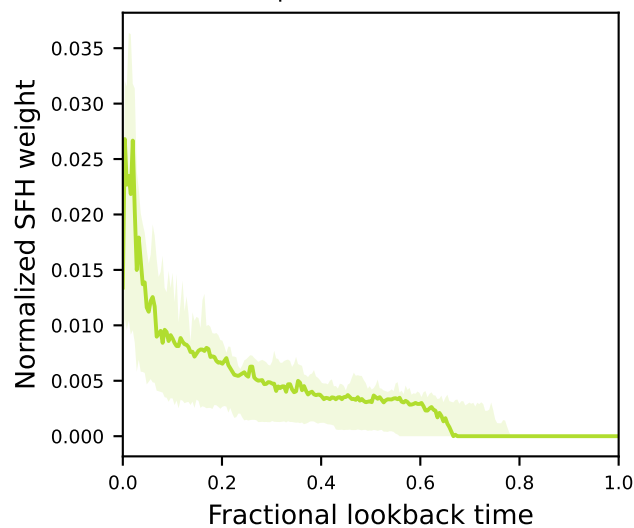
Complete candidate SFH



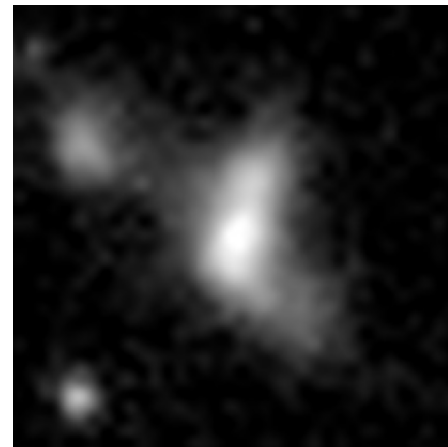
rank 3: 2634532333655237802
 $\log M^*=9.00$ (+0.00) $z=0.588$
 $D=0.458$



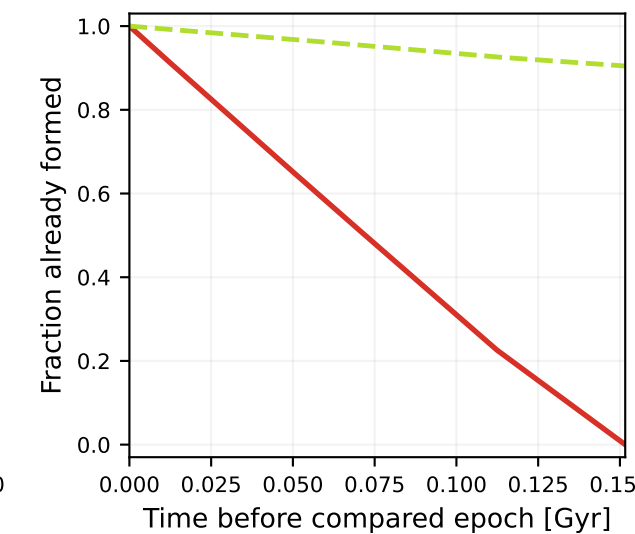
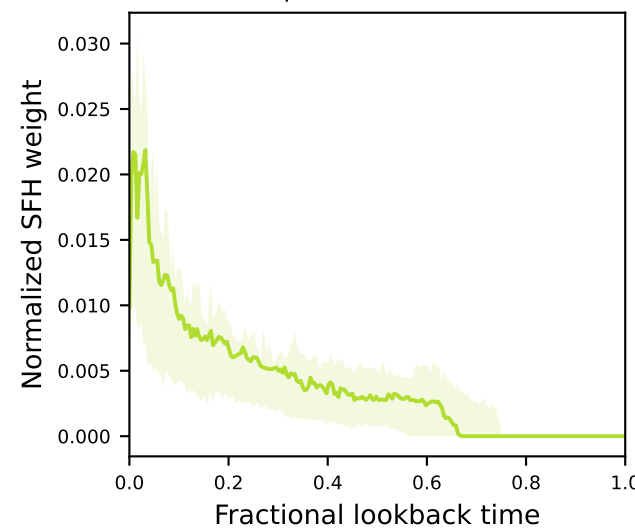
Complete candidate SFH



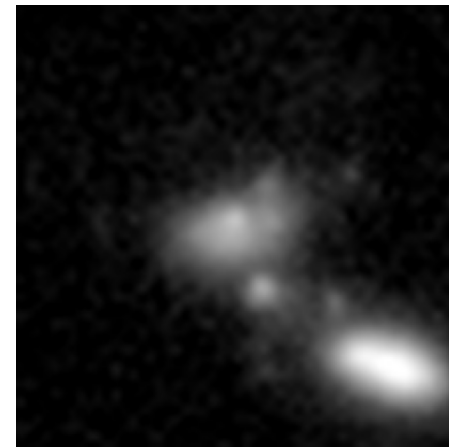
rank 4: 2765951524658897054
 $\log M^*=9.10$ (+0.10) $z=0.580$
 $D=0.470$



Complete candidate SFH



rank 5: 2639206210644703097
 $\log M^*=8.91$ (-0.09) $z=0.260$
 $D=0.470$



Complete candidate SFH

